# RE 3: CSTR study: The reaction of t-butyl chloride and water

TKP 4110 - Reaksjonssteknikk
13. september 2024

**Gruppe B.21**
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# 1 Theory

Short theoretical introduction to the reaction system. Think about: Which parameters must be measured and are necessary for calculations of conversion level, the rate constant and the activation energy? How can the inlet and outlet concentrations, conversion level and consumption rate of tBC be calculated?

## 1.1 How does the reaction work

When water and t-bytyl choride reacts they can follow these two equations

$$(CH_3)_3CCl + H_2O- > (CH_3)_3COH + HCl \tag{1.1}$$

$$(CH_3)_3CCl + H_2O- > (CH_3)_2CCH_2 + H_3O^+Cl^- \tag{1.2}$$

The first reaction, 1.1, is a substitution reaction, and the second, 1.2, is an elimination reaction. Both of these happens in the system and in competition of each other. The slow step both of these reactions is the generation of the carbocation making this a first order reaction. Because of this the rate expression can be written as

$$-r = kC_{t-BC}. \tag{1.3}$$

From this equation the $k$ is the reaction rate constant that can be found from the Arrheinus equation

$$k = Ae^{-E/RT}, \tag{1.4}$$

where $A$ is the pre-exponetial factor, $E$ is the activation energy, $R$ is the gass constant and $T$ is the temperature in kelvin.

## 1.2 How will the measured values be used to calculate kinetic parameters

The mass balance in a in a CSTR-system can be written as

$$\nu_f(C_i) - \nu_p(C_i)_p = -rV \tag{1.5}$$

Where $\nu$ is the volumetric flow rate of feed, and $V$ is the reaction volume. When steady-state comnditions can be assumed ($\nu_f = \nu_p = \nu_0$) the mass balance equation can be rewritten to

$$(C_i)_f - (C_i)_p = -r\frac{V}{\nu_0} = -r\tau. \tag{1.6}$$

Where $\tau$ is the mean residence time. When this expression is inserted in the expression for a first order reaction 1.3 it will look like

$$(C_i)_f - (C_i)_p = k(C_i)_p\tau, \tag{1.7}$$

which, when solved for the rate constant, can look like

$$k = \frac{1}{\tau}[\frac{(C_i)_f}{(C_I)p} - 1] \tag{1.8}$$

From the Arrheinus equation 1.4 the activation energy can be calculated

$$ln(k) = ln(A) - \frac{E}{R}(\frac{1}{T}) \tag{1.9}$$

### 1.3 plotting

When plotting equation 1.9 against $\frac{1}{T}$ it will give a linear graph with a slope of $\frac{-E}{R}$, this is where the activation energy is calculated. The pre-exponential factor for the Arrheinus equation can also be found from the plot.

## 2 Experimental

The experiment will be run at six different sets of operating conditions. The operation conditions are given in the table 2.1.

**Tabell 2.1:** Caption

| Run | Water bath temp. | Pump level |
|---|---|---|
| 1 | 35 | 80 |
| 2 | 35 | 60 |
| 3 | 40 | 80 |
| 4 | 40 | 60 |
| 5 | 45 | 80 |
| 6 | 45 | 60 |

For the experiment, one solution of $35M\mathrm{H_2O}$ in methanol, and one solution of $o.025M$ t-butyl chloride in methanol is prepared. To start the experiment, first the water bath temperature is set. Then the feed-hoses are connected to the reactant solutions. The magnetic stirrer is turned on, and the pH-meter is re-calibrated using the available buffer solutions. The pump-level is set. The pH-value, the elapsed time, and the reactor's temperature is recorded in the lab journal every three minutes as the reactor is approaching steady-state, at each experimental condition.

# A Calculations

## A.1 The amount of the components

Equation Likning (A.4) can be deduced from the equations A.1, A.2, A.3, and used to calculate the amount of each of the components.

$$n = C \cdot V_{total} \tag{A.1}$$

$$V = \frac{m}{\rho} \tag{A.2}$$

$$m = n \cdot M_m \tag{A.3}$$

$$V = \frac{C \cdot V_{total} \cdot M_m}{\rho} \tag{A.4}$$

The equation gives:

$$V_{(CH_3)_3CCl} = \frac{0.025\,\mathrm{mol/L} \cdot 92.6\,\mathrm{g/mol} \cdot 1000\,\mathrm{ml}}{0.851\,\mathrm{g/cm^3}} = 2.720\,\mathrm{ml} \tag{A.5}$$

$$V_{H_2O} = \frac{35\,\mathrm{mol/L} \cdot 18\,\mathrm{g/mol} \cdot 1000\,\mathrm{ml}}{1\,\mathrm{g/cm^3}} = 630\,\mathrm{ml} \tag{A.6}$$

For the t-butyl chloride solution in methanol, 997.28 mL of methanol is required, while for the water solution in methanol, 370 mL of methanol is needed.

## A.2 Expected total consumption of reactants and solvent

# B Example calculations

# C HSE

The chemicals and gasses used in this lab are methanol and t-butyl chloride. Both t-butyl chloride and methanol are highly flammable, they can cause eye and skin irritation and dizziness. T-butyl choride may lead to cancer after prolonged exposure. Methanol may cause nerve damage if absorbed into the body. The use of personal protective equipment such as protective gloves when handling the chemicals, along with a lab coat, goggles and closed toe shoes, is recommended. The use of an extractor hood when handling the chemicals is nessecary.

Gas detectors will alert in case of a leakage, and the lab assistant should be notified. Other potential risks include chipped or broken glass. Glass equipment may break and cause small cuts.